

Dylan Cashman

Visual Analytics and Data Science

Dylan Cashman. PhD.
415 South Street. Waltham, MA 02453
☎ 617-999-3634
✉ depcashman@gmail.com
dylancashman.github.io

Professional Appointments

- 08/23 – Present **Assistant Professor**, *Michtom School of Computer Science*, Brandeis University, Waltham, MA.
- 07/20 – 07/23 **Senior Expert, Data Science and Advanced Visual Analytics**, *Novartis Pharmaceutical Corporation*, Cambridge, MA, Data and Artificial Intelligence.
•Product owner for commercial project with 4 million USD annual budget
•Data scientist and visual analytics designer for safety + commercial projects
- 01/20 – 04/20 **Part-time Lecturer**, *Northeastern University*, Boston, MA, Khoury College of Computer Sciences.
- 05/18 – 08/18 **Research Intern**, *MIT IBM Watson AI Lab*, Cambridge, MA, Visual AI Group.
- 06/16 – 08/16 **Research Intern**, *Palo Alto Research Center*, Palo Alto, CA, System Sciences Lab.
- 06/15 – 08/15 **Research Intern**, *MIT Lincoln Laboratory*, Lexington, MA, Cyberanalytics and Decisions Systems.
- 09/10 – 08/14 **Development Manager**, *Annkissam*, Cambridge, MA.

Academic Degrees

- June 2020 **PhD in Computer Science**, *Tufts University*, Medford, MA.
Thesis: “Bridging the Human-Machine Gap in Applied Machine Learning with Visual Analytics”
Advisor: Remco Chang
- May 2016 **MSc in Computer Science**, *Tufts University*, Medford, MA.
Project: “Big Data, Bigger Audience: A Method for Adapting Statistical Methods for a Wider Audience of Users”
Advisor: Remco Chang
- May 2010 **ScB in Mathematics**, *Brown University*, Providence, RI.

Publications

Journal Publications

- [1] James Petullo, Sonny George, **Dylan Cashman**, and Nianwen Xue. Veccisc: Improving confidence-informed self-consistency with reasoning trace clustering and candidate answer selection. In *Findings of the Association for Computational Linguistics: ACL*, pages 26184–26200, 2026.
- [2] Areen Khalaila, Lane Harrison, Nam Wook Kim, and **Dylan Cashman**. “They Aren’t Built for Me”: An Exploratory Study of Strategies for Measurement of Graphical Primitives in Tactile Graphics. *IEEE Transactions on Visualization & Computer Graphics*, 32(01):1–12, January 2026.
- [3] **Dylan Cashman**, Mark Keller, Hyeon Jeon, Bum Chul Kwon, and Qianwen Wang. A critical analysis of the usage of dimensionality reduction in four domains. *IEEE Transactions on Visualization and Computer Graphics*, pages 1–20, 2025.
- [4] **Dylan Cashman**. PAC Learning Or: Why We Should (and Shouldn’t) Trust Machine Learning. *The Journal of Visualization and Interaction*, 2025.
- [5] Sonny George, Chris Sypherd, and **Dylan Cashman**. Probing the capacity of language model agents to operationalize disparate experiential context despite distraction. *Findings of the Association for Computational Linguistics: EMNLP*, 2024.

- [6] Ashley Suh, Gabriel Appleby, Erik W. Anderson, Luca Finelli, Remco Chang, and **Dylan Cashman**. Are Metrics Enough? Guidelines for Communicating and Visualizing Predictive Models to Subject Matter Experts. *IEEE Transactions on Visualization & Computer Graphics*, 30(07):4137–4153, July 2024.
- [7] Raea Rasmussen, David E Levary, Muna Akhtar, Chelsea S Crittle, Megan Gately, Jeremy Pagan, Andrea Brennen, **Dylan Cashman**, Alia N Wulff, Michael I Norton, Samuel I Sommers, and Heather L Urry. White (but not black) americans continue to see racism as a zero-sum game; white conservatives (but not moderates or liberals) see themselves as losing. *Perspectives on Psychological Science*, 2022.
- [8] Mateus Espadoto, Gabriel Appleby, Ashley Suh, **Dylan Cashman**, Mingwei Li, Carlos Scheidegger, Erik W Anderson, Remco Chang, and Alexandru C Telea. Unprojection: Leveraging inverse-projections for visual analytics of high-dimensional data. *IEEE Transactions on Visualization and Computer Graphics*, 2021. Accepted October 2021.
- [9] **Dylan Cashman**, Shenyu Xu, Subhajit Das, Florian Heimerl, Cong Liu, Shah Rukh Humayoun, Michael Gleicher, Alex Endert, and Remco Chang. Cava: A visual analytics system for exploratory columnar data augmentation using knowledge graphs. *IEEE Transactions on Visualization and Computer Graphics*, 27(2):1731–1741, 2020.
- [10] **Dylan Cashman**, Adam Perer, Remco Chang, and Hendrik Strobelt. Ablate, variate, and contemplate: Visual analytics for discovering neural architectures. *IEEE Transactions on Visualization and Computer Graphics*, 26(1):863–873, 2019.
- [11] **Dylan Cashman**, Shah Rukh Humayoun, Florian Heimerl, Kendall Park, Subhajit Das, John Thompson, Bahador Saket, Abigail Mosca, John Stasko, Alex Endert, Michael Gleicher, and Remco Chang. A user-based visual analytics workflow for exploratory model analysis. In *Computer Graphics Forum*, volume 38, pages 185–199. Wiley Online Library, 2019.
- [12] Subhajit Das, **Dylan Cashman**, Remco Chang, and Alex Endert. Beames: Interactive multimodel steering, selection, and inspection for regression tasks. *IEEE Computer Graphics and Applications*, 39(5):20–32, 2019.
- [13] **Dylan Cashman**, Genevieve Patterson, Abigail Mosca, Nathan Watts, Shannon Robinson, and Remco Chang. RNNbow: Visualizing learning via backpropagation gradients in RNNs. *IEEE Computer Graphics and Applications*, 38(6):39–50, Nov 2018.

Refereed Conference / Symposium Publications

- [14] Saad Hassan, Areen Khalaila, **Dylan Cashman**, Ather Sharif, and Rebecca Faust. A scoping review of human–computer interaction research on the accessibility of data visualizations: New technologies, procedures, and designs. *ACM ASSETS*, 2026.
- [15] Ashley Suh, Ab Mosca, Shannon Robinson, Quinn Pham, **Dylan Cashman**, Alvitta Ottley, and Remco Chang. Inferential tasks as an evaluation technique for visualization. *EuroVis Short Papers*, 2022.
- [16] Zhe Wang, **Dylan Cashman**, Mingwei Li, Jixian Li, Matthew Berger, Joshua Levine, Remco Chang, and Carlos Scheidegger. Neuralcubes: Deep representations for visual data exploration. In *IEEE BigData*, 2021.
- [17] Subhajit Das, **Dylan Cashman**, Remco Chang, and Alex Endert. Gaggle: Visual analytics for model space navigation. In *Graphics Interface*, 2020.
- [18] Abigail Mosca, Shannon Robinson, Meredith Clarke, Rebecca Redelmeier, Sebastian Coates, **Dylan Cashman**, and Remco Chang. Towards data science for the masses: A study of data scientists and their interactions with clients. In *EG/VGTC Conference on Visualization*, 2019.
- [19] Bob Price, Lottie Price, **Dylan Cashman**, and Marzieh Nabi. Efficient bayesian detection of disease onset in truncated medical data. In *Healthcare Informatics (ICHI), 2017 IEEE International Conference on*, pages 208–213, 2017.

Other Refereed Conference / Symposium / Workshop Publications

- [20] Thomas Breimer, Pirm Dhararag, Paras Pandey, Kiya Gedamu, Marie Amandine Reyes, Tal Spector, Galileo Steinberg, Nathan Subrahmanian, Paul Weir, Peter Williams, Weichen Zhao, and **Dylan Cashman**. Facilitating Reproducibility in Visualization Systems Research Through Containerization. In *BELIV Workshop at IEEE VIS*, 2026.

- [21] Pirm Dhararag and **Dylan Cashman**. MolExplain: An Interactive Tool for Explainable Molecular Property Prediction . In *Workshop on Visual Analytics in the age of Autonomous Science at IEEE VIS*, 2026.
- [22] Sonny George, Chris Sypherd, Rocco Ahching, and **Dylan Cashman**. Benchmarking Arbitrary Natural Language Tasks in 3D Open Worlds. In *Embodied AI Workshop (EAI) at CVPR*, 2025.
- [23] Ashley Suh, Gabriel Appleby, Erik W Anderson, Luca Finelli, and **Dylan Cashman**. Communicating performance of regression models using visualization in pharmacovigilance. In *Workshop on Visual Analytics in Healthcare at IEEE VIS*, 2021.
- [24] **Dylan Cashman**, Yifan Wu, Remco Chang, and Alvitta Ottley. Inferential tasks as a data-rich evaluation method for visualization. In *Evaluation of Interactive Visual Machine Learning Systems at IEEE VIS*, 2019.
- [25] **Dylan Cashman**, Adam Perer, and Hendrik Strobelt. MAST: A tool for visualizing CNN model architecture searches. In *Debugging ML Workshop at ICLR*, 2019.

Awards and Honors

- 2025 • Best Paper Award: IEEE VIS Full Papers (top 1%)
- 2023 • Best Submission Award: VISxAI Workshop for AI Explainability at IEEE VIS
- 2022 • Best Paper Award: EuroVis Short Papers Track
- 2018 • Best Paper Award: Symposium on Visualization in Data Science at IEEE VIS
• 3rd Place in 15 minute research talks at Tufts Graduate Research Symposium
- 2017 • Best Paper Award: Workshop on Visual Analytics for Deep Learning at IEEE VIS
- 2016-2018 • Provost's Fellowship, Tufts University

Conference Activities

- | | |
|-------------------|---|
| Organization | <ul style="list-style-type: none"> • Web Chair: IEEE VIS 2021-2024 • Papers Chair: Symposium on Visualization for Data Science (VDS) 2024
co-chair: Ana Crisan • Panel Organizer: VIS 2023 Panel on How should VIS4ML Redefine Itself in the Rapid Evolution of AI?
co-organizers: Junpeng Wang, Qianwen Wang |
| Session Chair | <ul style="list-style-type: none"> • Symposium on Visualization for Data Science (VDS) 2019, 2023 |
| Program Committee | <ul style="list-style-type: none"> • IEEE VIS Full Papers (VIS), 2022-2023, 2025-2026 • IEEE VIS BELIV Workshop, 2026 • Eurographics Eurovis, 2026 • IEEE Pacific VIS VisMeetsAI Workshop, 2022-2026 • IEEE VIS Short Papers (VIS), 2020-2021 • IEEE VIS Symposium on Visualization for Data Science (VDS), 2019-2020, 2023 • IEEE VIS Workshop on Visualization for AI Explainability (VISxAI), 2019-2020 |

Reviewing Activities

- | | |
|----------|--|
| Proposal | <ul style="list-style-type: none"> • National Science Foundation (NSF), 2026 |
| Journal | <ul style="list-style-type: none"> • IEEE Transactions on Computer Graphics and Visualization (TVCG), 2020-2026 • Computers and Graphics, 2026 • Computer Graphics and Applications (CG&A), 2025 • Computer Graphics Forum (CGF), 2020, 2023, 2024 • Artificial Intelligence Review, 2024 • The Visual Computer, 2023 • IEEE Transactions on Big Data (TBD), 2017-2018, 2022 • ACM Transactions on Interactive Intelligent Systems (TiiS), 2020-2021, 2024 |

- Distill.pub, 2018
- Conference
 - IEEE Conference on Visualization (VIS), 2018-2026
 - ACM Computer Human Interaction (CHI), 2018, 2020-2023, 2026
 - Eurographics Conference on Data Visualization (Eurovis), 2020, 2023
 - IEEE Pacific Visualization, 2018
 - ACM Computer Human Interaction (CHI) Late-breaking Work, 2015
- Workshop
 - Language Understanding in the Human-Machine Era at EMNLP (LUHME), 2026
 - IEEE Pacific VIS VisMeetsAI, 2022-2024
 - Visualization for AI Explainability (VISxAI), 2018-2020
 - Symposium on Visualization Data Science (VDS), 2019-2020, 2023
- Other
 - Mathematics of Information Technology and Complex Systems (MITACS) Accelerate Proposals, 2025
 - MITACS Elevate Proposals, 2023

Invited Talks

- November 8, 2025 Next Gen Data Science Day (Waltham, MA).
Panelist: Careers Session
- February 19, 2025 University of Utah (Remote Zoom Lecture). Visualization Seminar
Towards the Generation of Descriptive and Accessible Data Representations for Data Science and AI
- November 27, 2023 Brandeis University (Waltham, MA). Faculty Research Presentation
Improving the Impact of AI with Visual Affordances
- November 16, 2023 University of Victoria (Victoria, BC, Canada). Graduate Information Visualization Course
Improving the Impact of AI with Visual Affordances
- October 26, 2023 IEEE VIS VISxAI Lightning Talks (Melbourne, VIC, Australia). Workshop
What are the hard things that don't have to be so hard: Visual Affordances for AI
- August 10, 2023 Brandeis University Data Science Internal Internship Program (Waltham, MA). Symposium
Effective Communication of Data Science Results
- March 14, 2023 Brandeis University (Waltham, MA). Symposium
Information Visualization
- March 6, 2023 MIT Lincoln Laboratory (Lexington, MA). Symposium
Improving the Impact of AI with Visual Affordances
- January 31, 2023 Fairfield University (Fairfield, CT). Symposium
Improving the Impact of AI with Visual Affordances
- January 20, 2023 Baruch College (New York, NY). Symposium
Improving the Impact of AI with Visual Affordances
- January 9, 2023 The Roux Institute at Northeastern University (Portland, ME). Symposium
Improving the Impact of AI with Visual Affordances
- November 29, 2022 Bard College (Annondale-on-Hudson, NY). Lecture
Improving the Impact of AI with Visual Affordances
- October 16, 2022 Symposium on Large Data Analysis and Visualization at IEEE VIS (Oklahoma City, OK).
Early Career Research Lightning Talks
What Do Users Want From Big Data?
- February 2021 Washington University in St. Louis (St. Louis, MO).
Graduate-level Visual Analytics Guest Lecture: Machine Learning for Visualization
- January 28, 2021 Pacific Northwest National Labs (Richland, WA).
Bridging the Human-Machine Gap in Applied AI with Visual Analytics

December, 2020	Novartis Institute for Biological Research (Cambridge MA). Advanced Visual Analytics Forum <i>Value of Visual Analytics for Insights, Strategy, and Design</i>
September 27, 2019	Mathworks (Natick, MA). Deep Learning Tea <i>Model Selection for Data Scientists</i>
October 1, 2018	IEEE Visualization Conference (Berlin, Germany). Doctoral Colloquium <i>Visual Analytics for Model Selection</i>
March 2, 2018	Tufts University (Medford, MA). Tufts Graduate Research Symposium <i>RNNbow: Visualizing Learning via Backpropagation Gradients in Recurrent Neural Networks</i>
June 13, 2017	Tufts University (Medford, MA). Tufts REU Lecture Series <i>Color Spaces and Color Places</i>
February 24, 2017	Tufts University (Medford, MA). Tufts Graduate Research Symposium <i>Chasing Waldo: Implicit Recovery of User Behavior and Intent from User Interaction Logs</i>

Committee Member

PhD Thesis Committee (External)	Varun Ramesh Jois . Brandeis University. 2026 Thesis Advisor: Antonella DiLillo
	Thomas Willkens . Brandeis University. 2023 Thesis Advisor: Jordan Pollack
Undergraduate Thesis Committee (Advisor)	Areen Khalaila . Brandeis University. 2025 Thesis Advisor: Dylan Cashman

Mentoring

2025	Vedanshi Shah (Graduate Research Assistant) Agentic AI Visualization Research. Work submitted to TVCG 2026 After graduation: MS Student at Babson College
2025	Areen Khalaila (Undergraduate Research Assistant) Tactile Graphics Research. Work published in TVCG 2025 After graduation: PhD Student at Tulane University
2021	Ashley Suh (PhD Intern at Novartis) Regression Model Communication Research. Work published in TVCG 2023 After graduation: Technical Staff at MIT Lincoln Laboratory
2019-20	Charlie Caron (Master's Student at Tufts) Model Selection Research.
2018	Nathan Watts (Undergraduate at Tufts) Deep Learning Research. After graduation: Machine Learning Software Systems Engineer at MITRE
2017	Alan Luo (Choate Rosemary Hall High School) VALT Summer Research

Teaching

2026	• Interactive Machine Learning Systems, COSI 212b, Brandeis University (15 students)
	• Deep Learning, COSI 165b, Brandeis University (30 students)
2025-2026	• Introduction to Machine Learning, COSI 104A, Brandeis University (70 students)

- 2024-2026 • Human Computer Interaction, COSI 125A, Brandeis University (70 students)
- 2024 • Advanced Programming in Java, COSI 12B, Brandeis University (60 students)
- 2023-2026 • Information Visualization, COSI 116A, Brandeis University (90 students)
- 2023 • Data Structures and Algorithms, COSI 21A, Brandeis University (60 students)
- Spring 2020 • Information Presentation & Visualization, DS 4200, Northeastern University (60 students)
- Fall 2019 • Co-lecturer. Visual Analytics, COMP 150-04, Tufts University (35 students)

- Spring, Fall 2016 • Head Teaching Assistant. COMP 40: Machine Structure and Assembly Language Programming. Tufts University.
- Fall 2015 • Graduate Teaching Assistant. COMP 61: Discrete Math. Tufts University.
- Fall 2009 • Teaching Assistant. MATH 0200: Multivariate Calculus, Brown University
- Spring 2009 • Grader. MATH 0520: Linear Algebra, Brown University
- Fall 2008 • Teaching Assistant. MATH 0190: Calculus II, Brown University
- Fall 2007 • Teaching Assistant. MATH 0190: Calculus II, Brown University

Volunteering

- 2026 • Brandeis University International Business School. Member of faculty advisory board on strategic AI curriculum plan.
- 2025-2026 • Waltham Public Schools. Member of advisory board on vocational computer science curriculum.
- 2022-2023 • Lynn Vision Steering Committee. Selected to provide citizen feedback on city of Lynn, MA master plan
- 2017 • Student Volunteer at IEEE VIS 2017, Phoenix, AZ
- 2012-2013 • Founding Member, Curriculum Designer, and Lecturer at Railsbridge Boston, a semiannual workshop for underrepresented populations to learn web development.
- 2010-2011 • Stavros Outdoor Access, helping individuals with disabilities enjoy winter sports at ski track in Weston, MA

Updated October 1, 2026